Course Structure

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Module #① Introduction to deep learning (DL)

- Difference b/w ML, DL, AI, GenAI, LLM, Agentic AI, DS, RAG, NLP
- Use cases of DL in the industry
- (domain wise) ⟷ examples

Module #② Introduction to Neural Networks (ANN | NN)

(Artificial Neural Network)

- connecting the dots between human nervous system and NN } Intuition
- gradient descent algorithm (GDA) → Batch GDA, → Mini-batch GDA, → Stochastic GDA. } → write / code our own GDAs [leveraging maths]
- single-layer perceptron (SLP) ✓
- Multiple- " " (MLP) ✓ (write our own codes)
- Parameters in NN : weights and biases
- activation functions (AFs)

Intuition → Mathematics → code.

Mathematics → (API to share resources)

→ Building a NN model from scratch

Module #③ Neural Network Framework

- Introduction to TensorFlow
- Introduction to Keras

} [sequential modeling approach]

- Bonus: comparison between Keras & PyTorch.

- MNIST database : Handwritten Digits Recognition

Module # 4 Convolutional Neural Networks (CNNs)

- image and their attributes
- concept of filter (kernels)
- pooling concept
- batch normalization and dropout.

- CIFAR-10 dataset : object classification using CNN.

Module # 5 Natural language Processing (NLP)

Module # 6 Recurrent Neural Network (RNN)

Module # 7 Long Short Term Model (LSTM)

Module # 8 Miscellaneous & Bonus topics.

- learning from pre-built jobs
- How to have clear roadmap for getting jobs in Gen AI world.
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GDRIVE LINK:
https://drive.google.com/drive/folders/1iGMoSZF8W9LF5zDuyTzPuVr1JrjaQr5-